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Reviews

Emerging Influence of the Intestinal Microbiota during Allogeneic Hematopoietic Cell Transplantation: Control the Gut and the Body Will Follow



Melissa D. Docampo^{1,*}, Jeffery J. Auletta², Robert R. Jenq³

¹ Department of Immunology, Weill Cornell Graduate School of Medical Sciences, New York, New York

² Host Defense Program, Hematology/Oncology/BMT and Infectious Diseases, Nationwide Children's Hospital and Department of Pediatrics, The Ohio State University College of Medicine, Columbus, Ohio

³ Adult Bone Marrow Transplantation Service, Department of Medicine, Memorial Sloan-Kettering Cancer Center and Weill Cornell Medical College, New York, New York

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A B S T R A C T

The intestinal microbiota has many critical roles in maintaining gastrointestinal epithelial and gastrointestinal systemic immune homeostasis. This review provides insight into how allogeneic hematopoietic cell transplantation (HCT) and its associated complications and supportive care therapies affect the microbiota. Additionally, the review discusses how preservation and restoration of the microbiota might be advantageous in decreasing HCT-related morbidity and mortality.

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INTRODUCTION

For over 40 years, researchers have known that the commensal bacteria inhabiting our intestines, collectively termed the intestinal microbiota, are important modulators of the biology of hematopoietic cell transplantation (HCT). Clinical studies using strategies to suppress the intestinal microbiota to prevent acute graft-versus-host disease (GVHD) showed considerable promise early on but eventually failed to demonstrate consistent benefit. The concept of manipulating the microbiota to improve outcomes for patients was not forgotten, but the best means of doing so has remained unclear.

Over the years, advances in culture-independent techniques have led to an enhanced understanding of bacterial subtypes beyond conventional microbiologic culture techniques. Sequencing of the 16S rRNA gene was developed in the 1980s as a powerful phylogenetic tool [1]. Coupled with subsequent genetic methodologies, including in situ hybridization and PCR, 16S characterization allows rapid identification of bacterial isolates, including potential pathogens from clinical samples [2]. A more recent major advance has been the advent of high-throughput sequencing modalities, or “deep sequencing” methods, that allow characterization of the composition of mixed bacterial samples. Deep sequencing of 16S rRNA has yet to be used routinely in the clinical setting because of practical barriers to implementation, including time, cost, and complexity, although newer, more rapid techniques are being developed [3].

A role for the microbiota has thus been re-examined in relation to a variety of clinical outcomes, ranging from obesity and atherosclerosis, to allergies and asthma, and

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* Correspondence and reprint requests: Melissa D. Docampo, Weill Cornell Graduate School of Medical Sciences, 1300 York Avenue, Department of Immunology and Microbial Pathogenesis, New York, NY 10065.

E-mail address: med2013@med.cornell.edu (M.D. Docampo).

even to cancer development and autism. HCT has been no exception, with several groups recently uncovering important relationships between the microbiota and outcomes in HCT recipients. In this review we provide an update regarding our knowledge of the biology of intestinal homeostasis in relation to the microbiota and a summary of recent findings implicating cross-talk between the microbiota and intestinal immunity that appears to have a significant impact on outcomes after HCT.

INTESTINAL EPITHELIAL HOMEOSTASIS

The human gastrointestinal (GI) tract can harbor an estimated 100 trillion individual bacteria belonging to roughly 1000 species in any single individual, and 15,000 species of bacteria have been identified from human GI samples [4]. These bacteria exist in a symbiotic relationship with their host, and thus a critical function of the intestinal mucosa is to enforce immunologic tolerance to these bacteria while maintaining the potential to mount protective responses against pathogens when necessary.

The intestinal epithelium, composed of intestinal epithelial cells (IECs) connected by tight junctions, creates a physical and biochemical barrier to separate luminal organisms from intestinal tissues [5]. Intestinal epithelium is organized into small finger-like projections called villi, and in between villi are tubular invaginations called crypts. Intestinal epithelial stem cells, which give rise to all subsets of differentiated IECs, reside at the base of these crypts and are dependent on Wnt signaling. They can develop into specialized secretory IECs, including goblet cells, enteroendocrine cells, enterocytes and Paneth cells [6]. Goblet cells secrete mucins into the intestinal lumen to form a mucous layer that acts as a first line of defense against microbes [7]. Paneth cells are specialized cells found within the crypts that secrete a variety of antimicrobial peptides, including defensins, cathelicidins, and calprotectins. These have activity against a variety of microbes and function by compromising the integrity of microbial cell membranes [8]. A recently described antimicrobial peptide, the C-type lectin regenerating islet-derived protein III gamma (RegIII γ), is secreted by enterocytes in a MyD88-dependent manner and plays an important role in separating luminal bacteria from the intestinal epithelial surface [9,10].

The intestinal mucosa is in contact with harmless commensal bacteria as well as potential pathogens, thereby functioning to immunologically survey the intestinal lumen [11]. IECs monitor the tone of the lumen, recognizing a variety of microbial products in both antigen-independent and antigen-dependent manners to participate in coordinated immune tolerance or, alternatively, immune response [8]. IECs express pattern-recognition receptors, including Toll-like receptors, nucleotide-binding oligomerization domain (NOD)-like receptors, and retinoic acid-inducible gene 1 (RIG-I)-like receptors, for the recognition of microbial ligands or endogenous signals associated with pathogenesis to modulate cellular responses [12,13]. A subpopulation of IECs called microfold cells mediate sampling of antigens and microorganisms for presentation to the mucosal immune system [14]. Subepithelial macrophages also sample luminal contents through transepithelial dendrites [15]. Dysregulation of the surveillance mechanisms used by the intestinal mucosa can lead to the development of inflammatory bowel disease and other disorders [16].

Interestingly, many aspects of intestinal immune homeostasis fail to develop in the absence of the intestinal

microbiota. Germ-free mice, housed in sterile conditions, show extensive defects and impaired development of gut-associated lymphoid tissues and antibody production [17]. IECs in germ-free mice have altered patterns of microvilli formation and decreased rates of cell turnover compared with wild-type, leading to defective expansion of defensins and other antimicrobial peptides.

A variety of hematopoietic cells participates in intestinal homeostasis. Myeloid cells, including macrophages and dendritic cells (DCs), are important mediators of both tolerance and protection. Nonmigratory macrophages maintain close contact with IECs to phagocytose and mediate clearance of pathogens and commensal bacteria that traverse the epithelial barrier [18]. In response to commensal bacteria, IECs produce cytokines via pattern-recognition receptor signaling, promoting the development of DCs and macrophages with tolerogenic properties [8], including specialized transforming growth factor- β -producing CD103⁺CD11b⁺ DCs within the GI tract that can induce forkhead box P3 (Foxp3)-expressing regulatory T cell (Treg) expansion [19]. Upon recognition of bacteria, DCs carry antigenic material and live bacteria to secondary lymphoid tissues, including mesenteric lymph nodes and Peyer's patches, and present them to adaptive immune cells [18]. This induces the differentiation of Tregs as well as gut-homing properties on T cells for the recruitment of recirculating mature cells to the original site of antigen encounter at the intestinal lamina propria [20]. Indoleamine 2,3-dioxygenase (IDO), an enzyme expressed by DCs and macrophages, catalyzes the initial rate-limiting step in tryptophan degradation. IDO is induced during inflammation by inflammatory cytokines, such as IFN- γ , and inhibits T cell activation through the consumption of tryptophan and subsequent expansion of Tregs, aiding in intestinal homeostasis [21]. Germ-free mice exhibit reduced levels of IDO, suggesting a role of the microbiota in regulation of IDO [22].

T cells can also play an important role in intestinal immunity. Commensal bacteria mediate intestinal immune tolerance by producing short-chain fatty acids, including acetate, propionate, and butyrate, through the fermentation of undigested carbohydrates. These have been shown to induce colonic Tregs through up-regulation of gut-homing molecules [23] and Foxp3 [24]. A subset of bacteria from the order Clostridiales has been identified as important for induction of colonic Tregs [25,26], potentially by up-regulating transforming growth factor- β to support Foxp3 induction. In contrast, pathogen-associated stimuli cause inflammatory responses via IL-1 and IL-6 induction, resulting in subsequent Th1 and Th17 activation [27].

B cells are an additional arm of the immune system active within intestinal tissues. Commensals have recently been shown to regulate B cell development within intestinal lamina propria [28]. IEC secretion of cytokines induces B cell class switching to IgA in a T cell-independent and T cell-dependent manner. IgA produced by local plasma cells is transported by IECs across the epithelial barrier into the intestinal lumen to act as another important line of defense against microbes [8].

In addition to myeloid cells and B and T lymphocytes, the recently described innate lymphoid cells (ILCs) [29,30] are also important in intestinal immune homeostasis. ILCs are classified into 3 distinct populations (groups 1, 2, and 3) on the basis of the expression of specific transcription factors, cell surface markers, and ability to secrete particular cytokines. ROR γ t-expressing group 3 ILCs in the intestine have

been implicated in regulating commensal bacteria by limiting local and systemic chronic inflammation. An important function of these ILCs is to produce IL-22, an epithelial cell regenerative factor, in response to IL-23 produced by myeloid cells [31]. In this regard, cellular mechanisms activated by inflammation serve to protect intestinal epithelium by restoring immune tolerance at the epithelial interface. ILCs have also been found to produce Csf2 in response to macrophage-derived IL-1 β , resulting in DCs producing retinoic acid to promote Treg recruitment [32].

Interactions between components of the microbiota represent yet another line of defense against potential pathogens. Commensal bacteria importantly function to prevent overgrowth of potential pathogens [33]. Adding to the complexity of microbiota composition are host and environmental factors, which can modulate microbiota composition and include age [34], antimicrobial use [35], disease [36], inflammation [37], metabolites [38], stress [39], and diet [40], particularly malnourishment [41,42] (Figure 1).

EFFECTS OF HCT AND GVHD ON INTESTINAL HOMEOSTASIS

Pre-HCT conditioning typically consists of a combination of chemotherapeutic agents with or without total body irradiation and antibody-mediated lymphocyte depletion and is required to allow engraftment of allogeneic hematopoietic cells. However, conditioning is associated with off-target effects and often produces significant alterations in intestinal homeostasis (Figure 2). In murine models, an increase in total body irradiation dose correlated with compromised intestinal barrier function, resulting in elevated levels of lipopolysaccharide in the serum [43]. A similar phenomenon was confirmed in patients, with increases in intestinal permeability seen only after myeloablative conditioning and not reduced-intensity conditioning [44]. The presence of GVHD, in turn, further greatly compromises intestinal barrier function, indicating synergistic damage to the epithelium [43].

The primary mechanism driving loss of intestinal barrier function is thought to result from cell death in the rapidly proliferating crypt compartment of the intestinal epithelium, leading to impaired replenishment of the villus epithelium. Mild loss of barrier function leads to mucosal inflammation, which can manifest clinically early on as nausea and later as diarrhea and abdominal discomfort [45]. Recovery of the intestinal epithelium from injury is dependent on repopulation by intestinal stem cells, which have been found in murine models to be lost in the setting of GVHD, a potentially critical event that determines recovery from conditioning- and inflammation-mediated injury [46,47]. In these models, strategies to support epithelial cell recovery that have shown promise for reducing GVHD include the Wnt agonist R-spondin1 [46] and IL-22 [47].

Interestingly, the effects of GVHD on intestine-derived antimicrobial molecules appear to be molecule-specific. Although both α -defensins and the antimicrobial lectin RegIII γ in the steady state are expressed by Paneth cells (and RegIII γ by enterocytes) [48], the α -defensins are profoundly reduced with GVHD [49], whereas RegIII γ is markedly up-regulated [50]. The human homologue of RegIII γ , RegIII α , is similarly found to be elevated in the serum of patients with intestinal GVHD [51]. How levels of antimicrobial compounds are modulated by GVHD may depend on expression patterns of the molecules.

IgA levels are also frequently reduced for many months after HCT. Reduced IgA levels in the intestinal lumen of mice can be seen after HCT [52] and in the serum of patients during the first 6 months following HCT, after which patients generally begin to recover normal IgA levels. Those who develop acute or chronic GVHD, however, remain IgA deficient [53]. In addition to impaired recovery of plasma cell function, GVHD is associated with perturbations in additional aspects of intestinal homeostasis. IL-22-producing ILCs have been found to be profoundly depleted in the intestines of mice with GVHD [47], which may contribute to loss of intestinal epithelial stem cells. Notably, Paneth cells

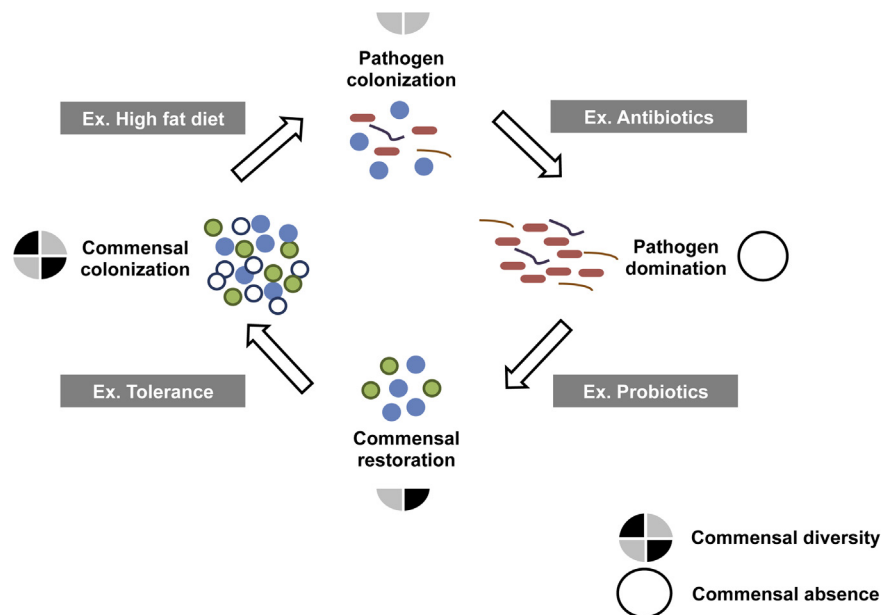


Figure 1. Fluctuations in intestinal microbial diversity reflect host and environmental factors. Commensal bacteria prevent overgrowth of potential pathogens. Host and environmental factors can modulate microbiota composition. A high-fat diet can lead to pathogen colonization, antibiotic use can lead to pathogen domination, use of probiotics restores commensal populations, and excessive tolerance leads to commensal overgrowth and increased commensal diversity.

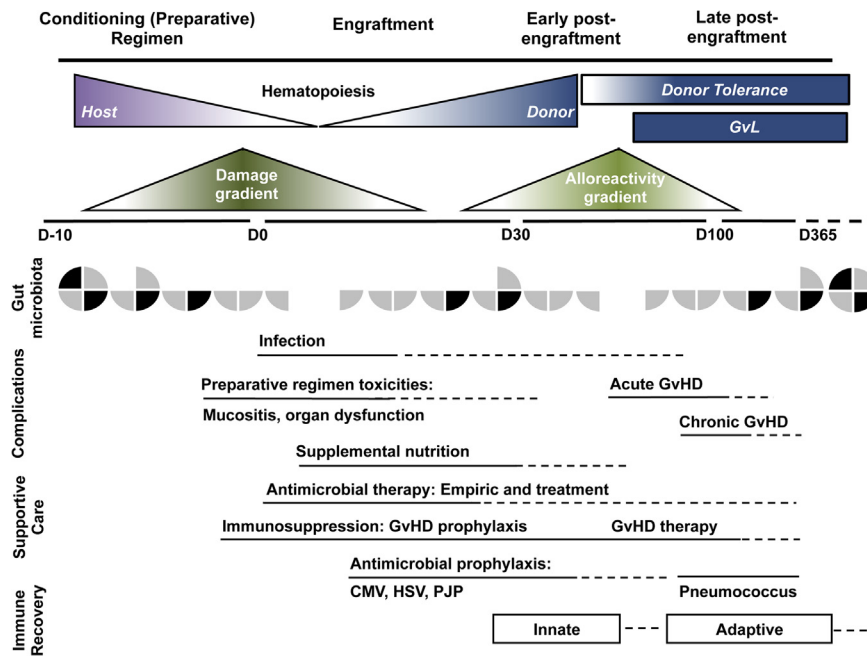


Figure 2. Transplant-related influences on the intestinal microbiota. Pre-HCT conditioning is required to allow engraftment of allogeneic hematopoietic cells. However, conditioning is associated with off-target effects like reduction in the amount and diversity of the microbiota, mucositis, organ dysfunction, and infection. Supportive care at this stage includes supplemental nutrition, antimicrobial therapy, and also GVHD prophylaxis. Antimicrobial prophylaxis could prevent cytomegalovirus (CMV), herpes simplex virus (HSV), and *Pneumocystis jiroveci* pneumonia (PJP) infection. Once engraftment is ongoing, the presence of GVHD further compromises intestinal barrier function, indicating synergistic damage to the epithelium, and therapy is necessary to complement innate and adaptive immune responses. GvL indicates graft-versus-leukemia.

are markedly reduced in the setting of GVHD in both murine models [49,52] and patients [54]. Reduction in Paneth cells was found to be prognostic for a low likelihood of response to treatment and increased nonrelapse mortality [54].

EFFECTS OF HCT AND GVHD ON INTESTINAL MICROBIOTA

Effects of GVHD on intestinal microbiota composition have been examined by several groups. Murine experiments have demonstrated that although HCT itself produces only mild changes in microbiota composition, onset of GVHD is associated with major fluctuations, including a generalized loss of species diversity, an expansion of members of the bacterial orders Enterobacteriales (including *Escherichia*, *Klebsiella*, and *Enterobacter* species) and Lactobacillales (including *Lactobacillus*, *Enterococcus*, and *Streptococcus* species), and a corresponding loss of obligately anaerobic bacteria from the phylum Firmicutes, including members of the order Clostridiales [49,52,55].

In contrast to mice, HCT alone is often associated with major changes in microbiota composition [56]. This difference is attributed to the frequent use of antibiotics in patients undergoing HCT, whereas antibiotics are not routinely used in murine experiments. During the course of hospitalization, many patients develop loss of flora diversity and concomitant oligoclonal expansions of certain bacteria. Common expanders include *Enterococcus*, *Streptococcus*, and various Enterobacteriales. Interestingly, expansion of these bacteria often precedes bloodstream infection by the same organisms [57], a previously underappreciated aspect of the pathophysiology of neutropenic sepsis in HCT patients. Perhaps the most significant risk factor for developing enterococcal expansion that could be identified was exposure to metronidazole during the hospital course [56],

suggesting that damage to the obligately anaerobic subset of commensals led to impaired suppression of *Enterococcus* resulting in outgrowth.

Patients who develop GVHD after allogeneic HCT display a characteristic shift from Clostridiales to Lactobacillales/Enterobacteriales [52,58,59] in a manner very similar to that seen in murine models. The etiology for these shifts in intestinal flora composition is unclear but may be related to changes in nutritional intake. A pattern of loss of members of Clostridiales, including *Roseburia*, *Faecalibacterium*, *Ruminococcus* and *Blautia* species, can be observed in volunteers placed on high-protein and low-carbohydrate diets [60] or on diets derived entirely from animal products [61]. Many of these Clostridiales perform the bulk of fermentation of consumed no digestible carbohydrates, and their metabolites are increasingly thought to produce health benefits [62], including induction of Tregs as described above. Thus, it is possible that the loss of Clostridiales observed in the setting of GVHD is maladaptive and may further contribute to inflammation within the intestine.

EFFECTS OF MICROBIOTA ON GVHD

Early murine studies from the 1970s found that broadly targeting the microbiota improved outcomes in murine models after allogeneic HCT. Mice transplanted in germ-free conditions [63] or receiving gut-decontaminating antibiotics [64] developed significantly less GVHD, demonstrating that the microbiota contribute to the development of GVHD-related lethality. In the 1980s, clinical trials of near-total bacterial decontamination with skin cleansing, oral antibiotics, and protective isolation were initially reported to result in improved outcomes [65,66]. However, subsequent clinical studies demonstrated no clear benefit [67–69]. The reasons behind these inconsistent results remain unclear but could

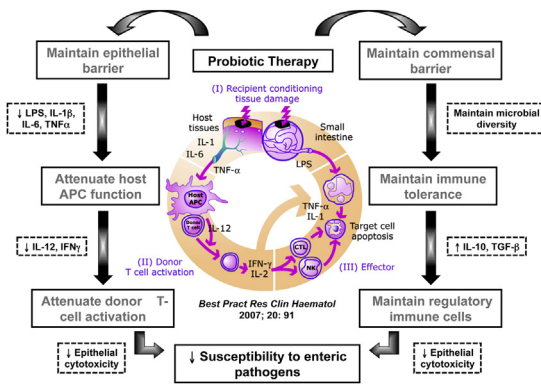


Figure 3. Putative roles for probiotic therapy in preventing acute GVHD. Probiotic therapy reduces susceptibility to enteric pathogens through the maintenance of the epithelial and commensal barriers. LPS indicates lipopolysaccharide; TNF- α , tumor necrosis factor- α ; APC, antigen presenting cell; TGF- β , transforming growth factor- β .

be due to variable success in total decontamination [70]; therefore, this expensive approach was abandoned in the early 1990s [71].

How the microbiota can modulate GVHD-mediated inflammation is an area of ongoing study. The bacterial product, lipopolysaccharide, which signals through TLR4, was identified to be an important proinflammatory mediator of GVHD [43]. The role of Toll-like receptors and MyD88/TRIF signaling in general has been investigated by several groups and remains controversial, with disparate results likely dependent on specific model differences as well as the tissue type where the role of signaling is being studied [72–74]. An additional pathway mediating inflammatory responses to the microbiota in the setting of GVHD includes the Nlrp3 inflammasome via production of IL-1 β [75]. Recently, bacterial translocation into intestinal tissues after total body

irradiation in mice was found to lead to recruitment of neutrophils into intestinal tissues [76]. This neutrophilic infiltration mediated localized tissue damage via production of reactive oxygen species, resulting in exacerbated GVHD. Yet not all bacterial sensing pathways are necessarily proinflammatory. For example, NOD2, which functions to recognize bacterial peptidoglycan, appears to mediate protective effects against GVHD in both mouse models [77] and humans [78].

Efforts to identify specific pro- and anti-inflammatory members of the microbiota have found that members of Enterobacteriales and *Enterococcus* may be potential contributors to increased GVHD in both mice [49,52] and humans [58,59], although these findings are based on associations, and causation as well as potential mechanisms remain to be elucidated. Murine studies have also found that certain intestinal lactobacilli can reduce experimental GVHD [52,79]. Thus, replenishing the microbiota through probiotic therapy may potentially offer a novel approach to attenuating GVHD and its associated risk for bloodstream infections (Figure 3). More severe neutropenia after nonmyeloablative conditioning has been found to be associated with increased GVHD and nonrelapse mortality [80]; whether neutropenia contributes directly to GVHD or alternatively is associated with increased use of antibiotics, which then leads to increased GVHD, remains unclear. Notably, neutrophil recruitment into the GI tract and subsequent oxygen radicals may amplify tissue cytotoxicity [76].

Recently, our group reported an analysis of patient flora samples early after HCT [81]. Eighty recipients of allogeneic HCT provided fecal samples at the time of neutrophil recovery, and microbial diversity, quantified by the inverse Simpson index, was predictive of overall survival. Mortality was particularly increased in patients with the lowest intestinal diversity, with overall survival rates at 3 years of 36% in the low-diversity group, compared with 60% and 67% in

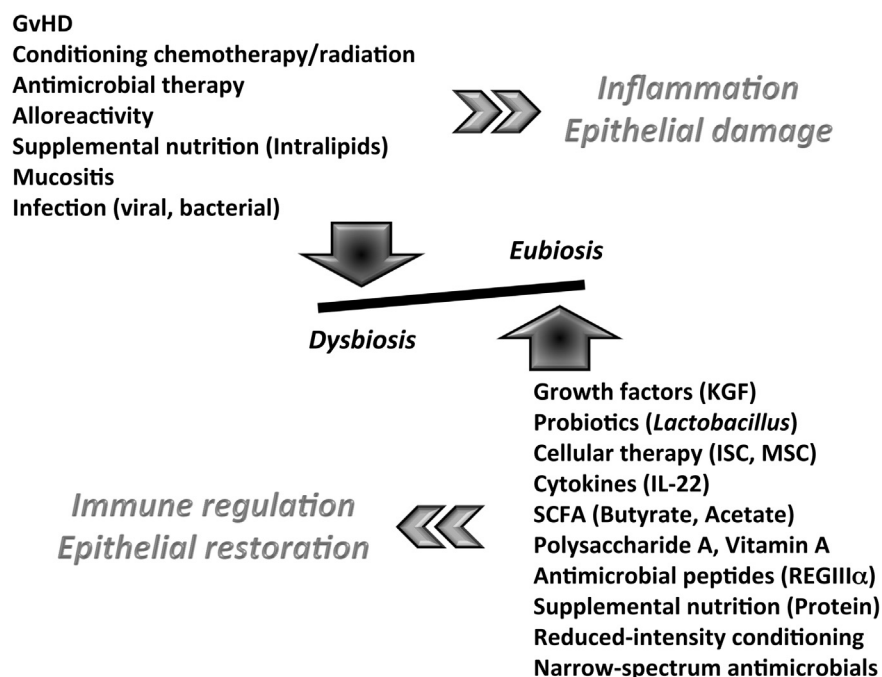


Figure 4. Microbiota homeostasis during alloHCT and proposed therapies targeting restoration in intestinal microbial diversity. A number of factors produce dysbiosis leading to inflammation and epithelial damage and are balanced by factors that produce eubiosis leading to immune regulation and epithelial restoration. KGF indicates keratinocyte growth factor; ISC, intestinal stem cells; MSC, mesenchymal stem cell; SCFA, short-chain fatty acid.

the intermediate- and high-diversity groups, respectively. The increase in mortality associated with lower diversity appeared to be largely attributed to increased death due to GVHD or infection rather than relapse or disease progression.

Whether the microbiota of the allogeneic HCT donor could impact on outcomes after allogeneic HCT is an interesting question that is not yet fully answered. Mouse models indicate that hematopoietic grafts derived from mice raised in germ-free conditions caused GVHD similar in severity to grafts from conventional mice, indicating that the presence or absence of donor microbiota does not appear to impact on GVHD [82]. Whether more subtle changes in the composition of the donor microbiota can modulate GVHD severity remains an open question.

FUTURE STRATEGIES

Recent work has resulted in a heightened appreciation for the strength of a bidirectional relationship that exists between the intestinal microbiota and the patient at risk for developing GVHD. Much work remains to better characterize precisely the bacterial mediators of increased or reduced risk for GVHD, bacterial ligands and metabolites that modulate host tissues, and the cellular components that carry out microbiota effects to change inflammatory tone (Figure 4). These efforts should pave the way to targeting the microbiota in a rational way to significantly mitigate the risk of developing GVHD in a manner that does not result in diminished global immune function or blunting of graft-versus-tumor effects that are critical to prevent malignant relapse.

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